

Maven – Interview Notes

1) What is Maven?

Maven is a Project Management and Build Automation Tool used mainly for Java projects. It helps manage project build lifecycle, dependencies, plugins, and documentation using pom.xml.

2) Build Lifecycle

Compile → Test → Package → Install → Deploy

Compile: Converts source code into bytecode.

Test: Runs unit tests.

Package: Creates JAR/WAR file.

Install: Stores in local repository.

Deploy: Uploads to remote repository.

3) Dependency Management

Dependencies are external libraries like MySQL Connector or Hibernate JAR files.

Transitive Dependencies

Some libraries depend on other libraries. Maven automatically downloads all required dependencies.

Version Compatibility

Maven helps manage compatible versions between frameworks like Spring and Hibernate.

4) Maven Advantages

Automatic dependency management, version control, build automation, and plugin support.

5) Plugins

Compiler Plugin, Surefire Plugin, WAR Plugin help automate tasks.

6) Alternatives

Gradle and Apache Ivy are alternatives to Maven.

Interview Summary

Maven is a build and dependency management tool that simplifies project development by automating build lifecycle, handling dependencies, and ensuring compatibility.